# ROS 2 Interface Guide for RFT Series (EtherCAT Model)

This guide provides instructions for interfacing the **RFT sensor**, which uses **EtherCAT communication**, within a **ROS 2** environment.

Steps 2 through 4 require a total of **three terminals** .
Each step must be executed in a **separate terminal**.

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## System Requirements

- ROS 2 must be installed
- Verified in a virtual environment using **Ubuntu 20.04** with **ROS 2 Foxy**

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# 1. ROS 2 Workspace Setup and Package Build

## 1-1) Create your workspace

- Create a directory that will contain your ROS 2 workspace.

  ```
  mkdir -p ~/{your_workspace}
  ```

  ```
  cd ~/{your_workspace}
  ```

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## 1-2) Create the 'src' folder

- The 'src' folder is not generated automatically. You need to create it manually to store ROS 2 packages.

  ```
  mkdir src
  ```

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## 1-3) Unzip your ROS 2 package into the 'src' folder

- Copy 'rft_ethercat_ros2_interface.zip' into 'src' and extract it

  ```
  cp ~/Downloads/rft_ethercat_ros2_interface.zip ~/{your_workspace}/src/
  ```

  ```
  cd ~/{your_workspace}/src
  ```

```
unzip rft_ethercat_ros2_interface.zip
```

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## 1-4) Build the Workspace

- Build all packages in the 'src' folder using 'colcon'.

```
cd ~/{your_workspace}
colcon build
```

- After the build completes, verify that the following directories are created:
  - build/
  - install/
  - log/

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# 2. EtherCAT Environment Setup and Interface Execution

- This section describes the complete process required to install dependencies, configure permissions, and run the EtherCAT interface

## 2-1) Install EtherCAT Python Dependency (pysoem)

```
sudo pip3 install pysoem
```

- Root privileges are required because EtherCAT uses **raw network sockets**.
- Setting network capabilities (**cap_net_raw**) also requires **sudo**.

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## 2-2) Configure Python Permissions for EtherCAT

- Check the Python version in use and grant raw network socket permission.

```
cd /usr/bin
ls python3.8
sudo setcap cap_net_raw+ep /usr/bin/python3.8
```

- Applying setcap restricts environment variables.
- This may prevent ROS 2 shared libraries from loading correctly.
- **Therefore, the interface must be executed as root.**
- A permanent non-root solution is under investigation.

## 2-3) Run the EtherCAT Interface Node

- Switch to the root user:

  sudo su

- Navigate to the workspace and source the ROS 2 environment:

  cd ~/{your_workspace}

  source /opt/ros/<your_ros2_distro>/setup.bash

  source ./install/setup.bash

- Move to the script directory and run the interface:

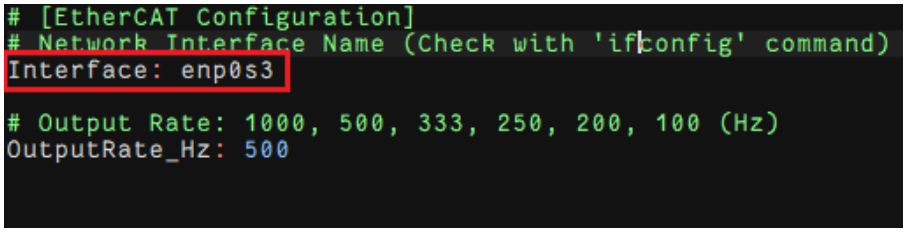
  cd ~/{your_workspace}/src/rft_ethercat_ros2_interface/script

  python3 interface.py

- Find the network adapter name using:

  ifconfig

- Identify the active network adapter used for EtherCAT communication (for example: **enp0s3**).
- In the script directory, open the config.txt file and enter the identified adapter name as shown below
- Replace enp0s3 with the actual adapter name detected on your system.



# 3. Check and Display FT Sensor Data

## 3-1) Open a New Terminal (Root Required)

```
sudo su
cd ~/{your_workspace}
source /opt/ros/<your_ros2_distro>/setup.bash
source ./install/setup.bash
```

---

## 3-2) Display FT Data

- Display the FT (Force/Torque) data published by the sensor:

```
ros2 topic echo /RFT/FT_data
```

```
    nanosec: 813804903
  frame_id: RFT
wrench:
  force:
    x: 1.6875948905944824
    y: 4.810939311981201
    z: 30.02758026123047
  torque:
    x: 0.08039180189371109
    y: -0.12530365586280823
    z: 0.002617656486108899
---
```

- This command continuously outputs FT sensor data to the terminal.
- Only FT data messages are displayed.

# 4. Sensor Control via ROS 2 Services

- Sensor control commands are sent using **ROS 2 services**.
- A service client is required to send commands to the sensor.

## 4-1) Launch the Service Client (Root Required)

- Open a new terminal and source the environment:

  sudo su

  cd ~/{your_workspace}

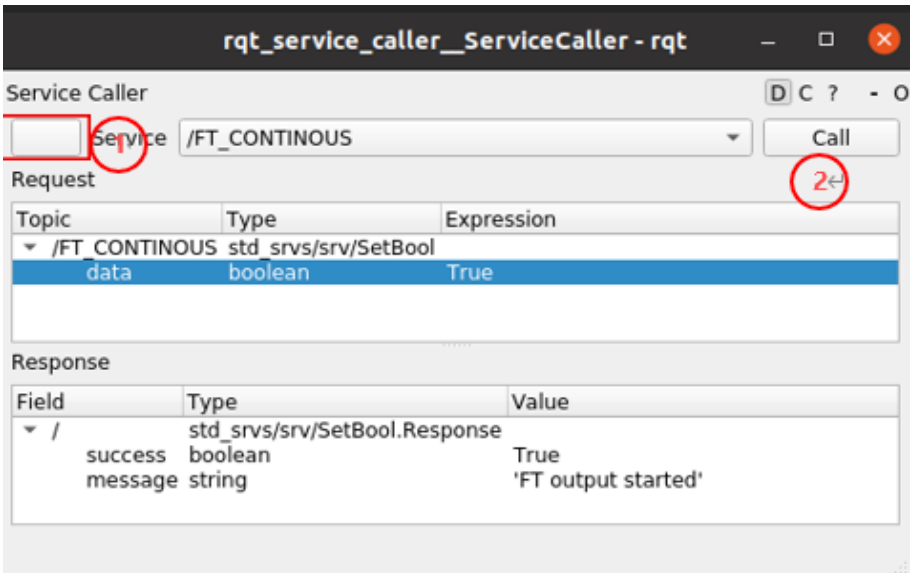
  source /opt/ros/<your_ros2_distro>/setup.bash

  source ./install/setup.bash

- Launch the service client using rqt_service_caller:

  ros2 run rqt_service_caller rqt_service_caller

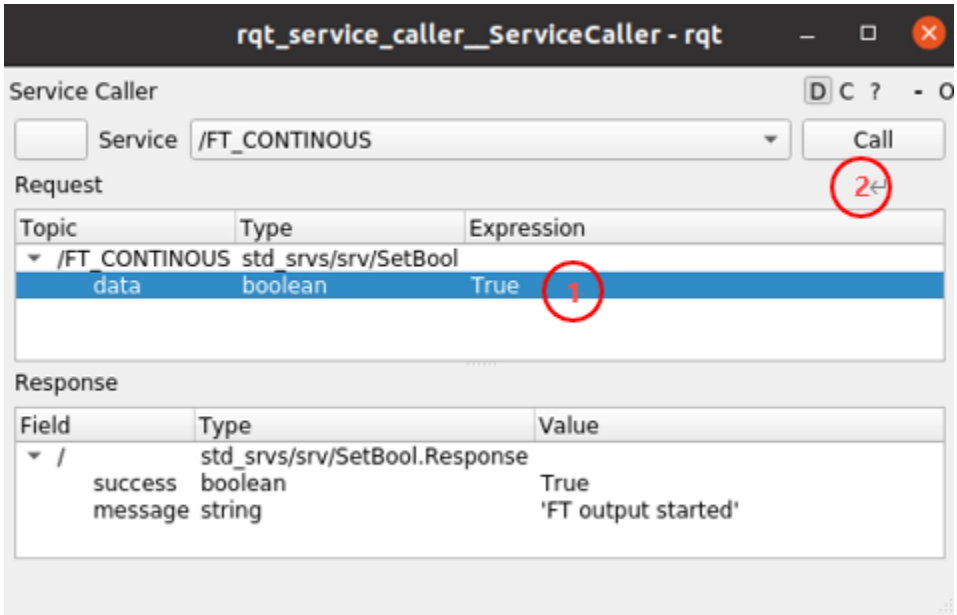
- Once the window opens, click the Update button (1), then select the desired service from the list (2).




- Before using any service other than FT output, /FT_CONTINOUS must be set to '**False**'.
- Other functions will not operate correctly while FT data output is active.
- Output messages from read/set services appear in the terminal used in **Section 2-3**.
- Start/Stop FT output messages appear in the terminal used in **Section 3-2**.
- Parameters returned by read/set services follow the tables in Sections **4-11**.

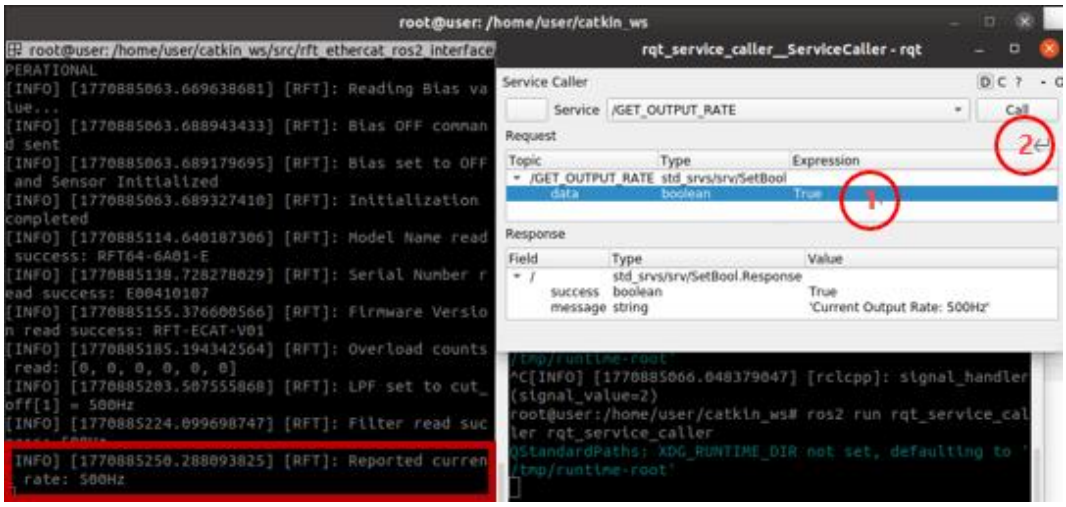
## 4-2) Stop / Start Data Output

- To stop FT data output:
  - Set the **expression** parameter of / FT_CONTINOUS to '**False**' (1)
  - Click **Call** (2)
- To start FT data output:
  - Set the parameter to '**True**' (1)
  - Click **Call** (2)



## 4-3) Read Data Output Rate

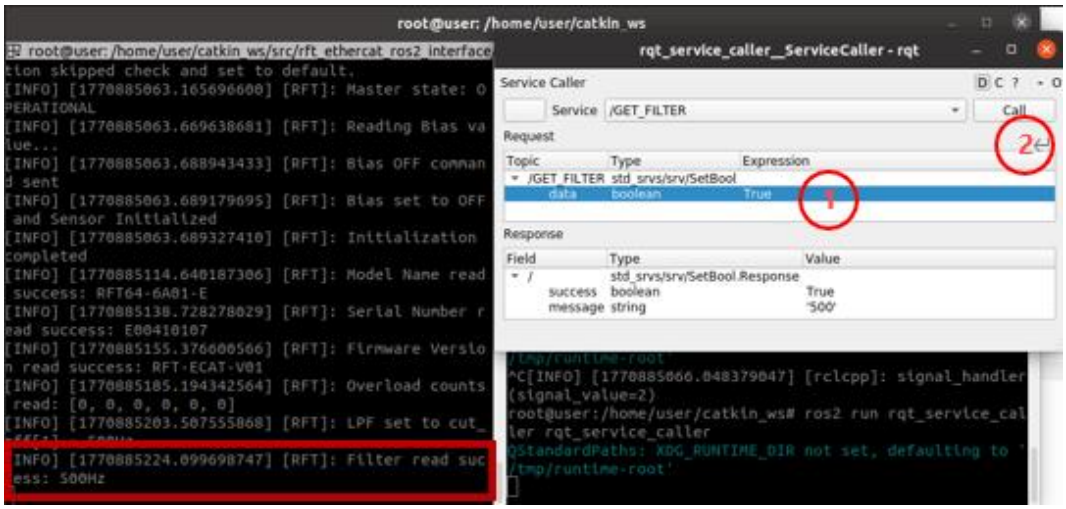
- Select the /GET_OUTPUT_RATE service.
- Set the **expression** parameter to '**True**'(1).
- Click **Call** (2).




- The current data output rate parameter is displayed.

## 4-4) Read Filter Setting

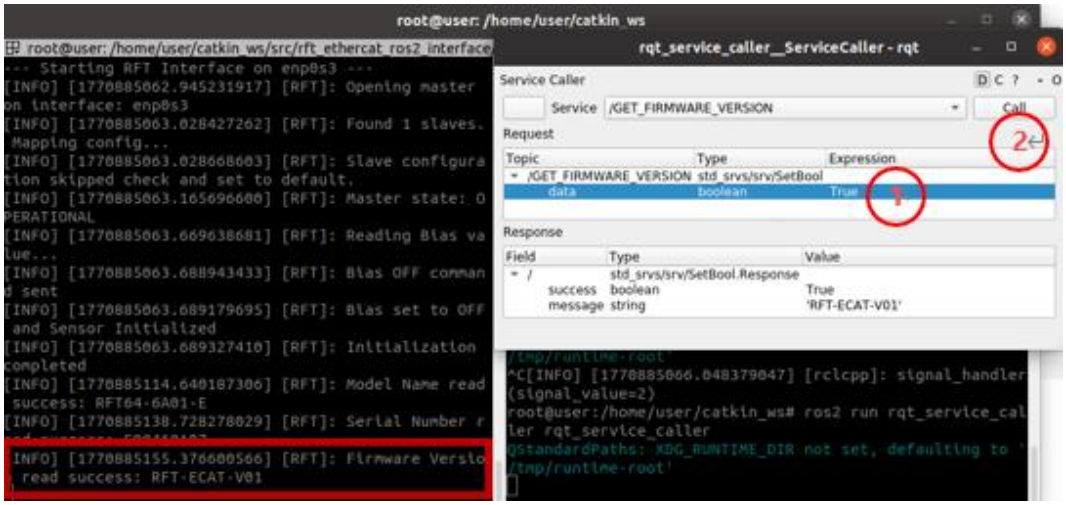
- Select the /GET_FILTER service.
- Set the **expression** parameter to '**True**'(1).
- Click **Call** (2).




- The current filter setting parameter is displayed.

## 4-5) Read Firmware Version

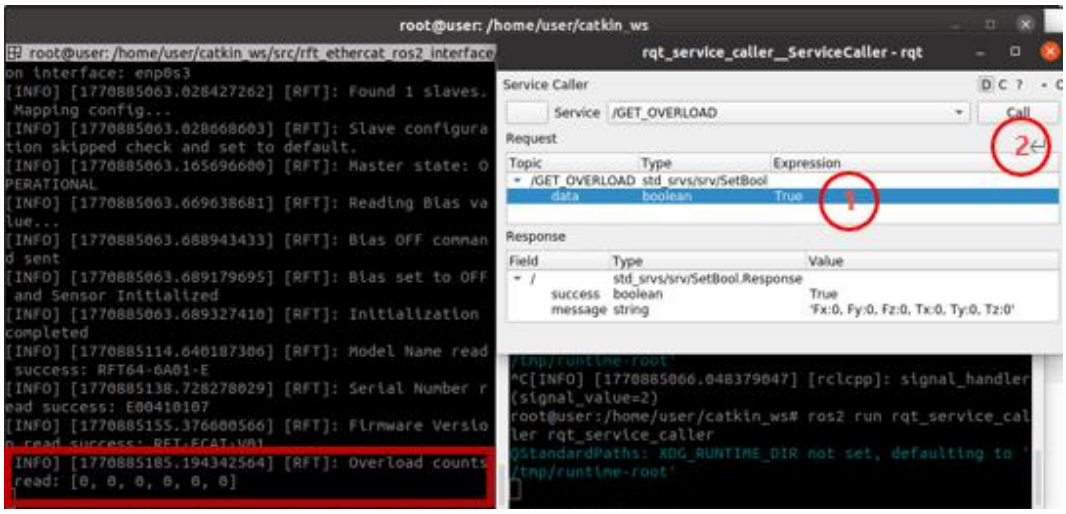
- Select the /GET_FIRMWARE_VERSION service.
- Set the **expression** parameter to '**True**' (1).
- Click **Call** (2).




- The firmware version is displayed.

## 4-6) Read Count of Overload Occurrence

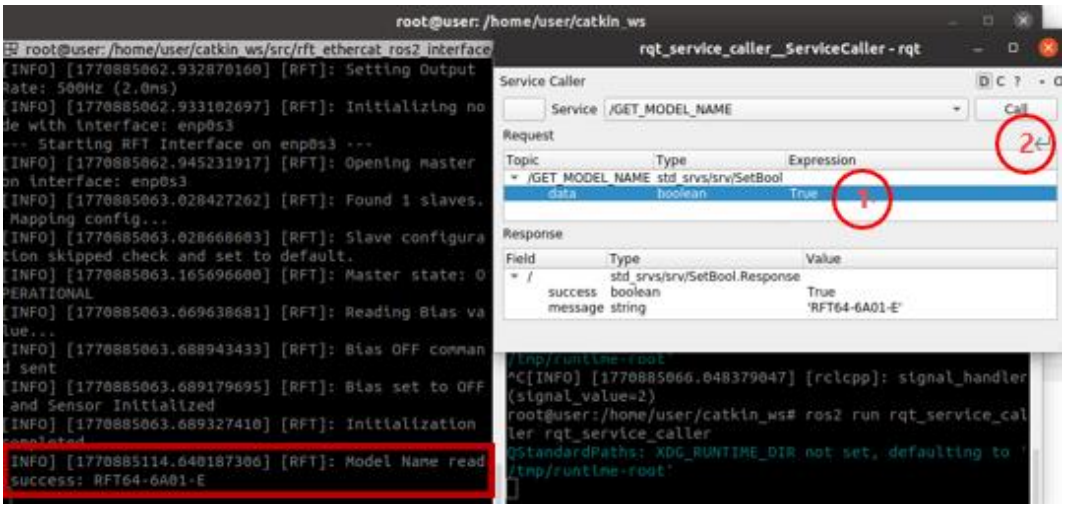
- Select the /GET_OVERLOAD service.
- Set the **expression** parameter to '**True**' (1).
- Click **Call** (2).




- Overload count values are displayed in the following order:
  - Fx, Fy, Fz, Tx, Ty, Tz

## 4-7) Read Model Name

- Select the /GET_MODEL_NAME service.
- Set the **expression** parameter to '**True**' (1).
- Click **Call** (2).




- The sensor model name is displayed.

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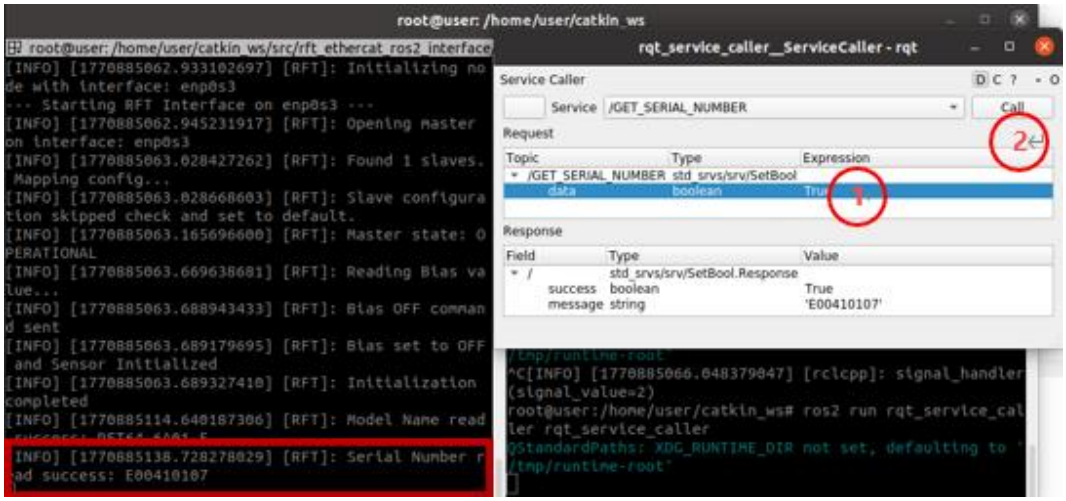
## 4-8) Read Serial Number

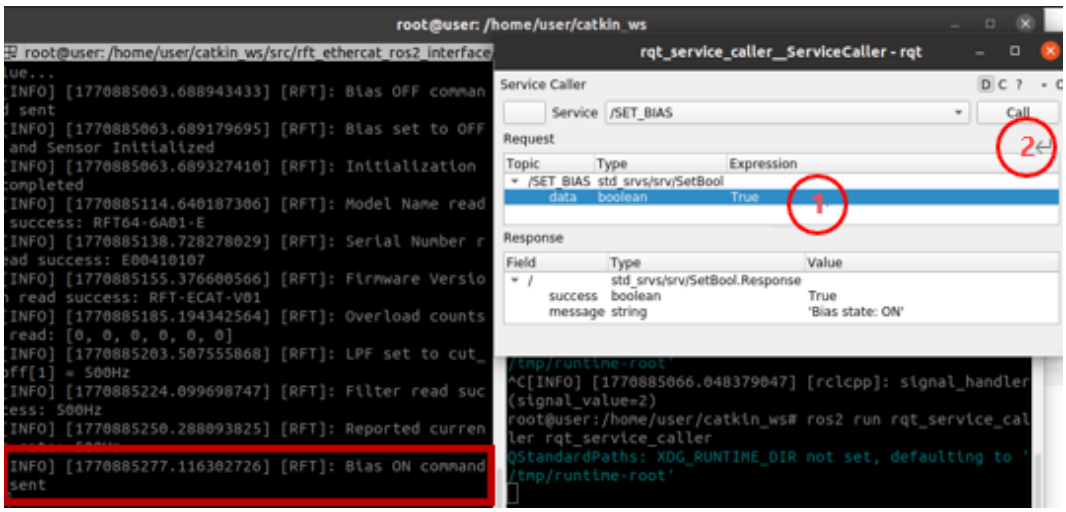
- Select the /GET_SERIAL_NUMBER service.
- Set the **expression** parameter to '**True**' (1).
- Click **Call** (2).




- The sensor serial number is displayed.

## 4-9) Set Bias

- To disable bias:
  - Set /SET_BIAS to '**False**' (1).
  - Click **Call** (2).
- To enable bias:
  - Set the parameter to '**True**' (1).
  - Click **Call** (2).



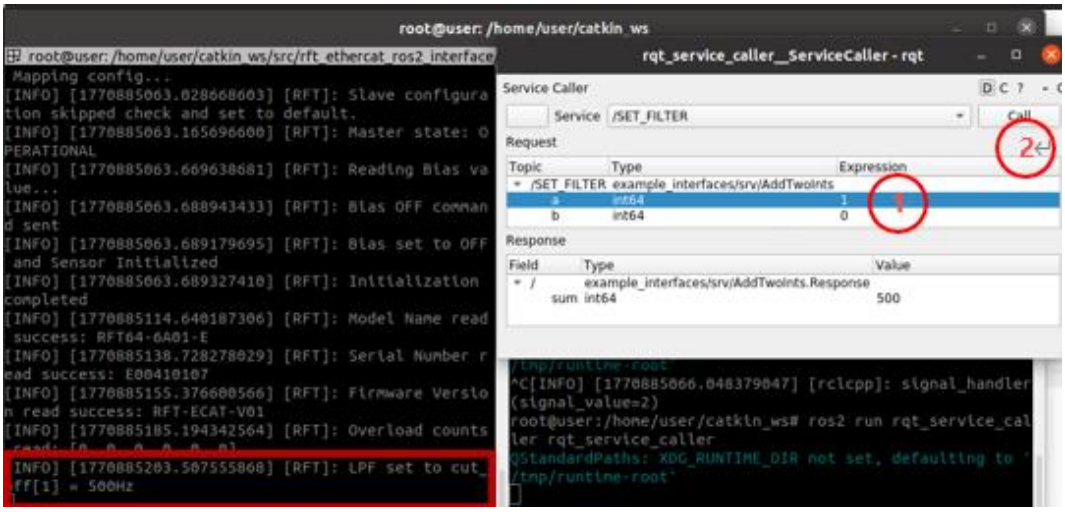

## 4-10) Set Data Output Rate

- The data output rate can be configured in the config.txt file located in the script directory.
- Change only the numeric value of the output rate to one of the values listed on the 5th line.
- After saving 'config.txt', **restart the Python interface executed in Step 2-3** for the change to take effect.

```
1 # [EtherCAT Configuration]
2 # Network Interface Name (Check with 'ifconfig' command)
3 Interface: enp0s3
4
5 # Output Rate: 1000, 500, 333, 250, 200, 100 (Hz)
6 OutputRate_Hz: 500
```

## 4-11) Set Filter

- Select the /SET_FILTER service.
- Enter the filter parameter value (not the filter type) in **request.a** (1).
- Click **Call** (2).




- Refer to the table below for filter parameters and cutoff frequencies:

| Filter Type | Filter Parameter | Cutoff Frequency [Hz] |
|---|---|---|
| 0(0x00) | 0(0x00) | No filter |
| 1(0x01) | 0(0x00) | No filter |
| 1(0x01) | 1(0x01) | 500 |
| 1(0x01) | 2(0x02) | 300 |
| 1(0x01) | 3(0x03) | 200 |
| 1(0x01) | 4(0x04) | 150 |
| 1(0x01) | 5(0x05) | 100 |
| 1(0x01) | 6(0x06) | 50 |
| 1(0x01) | 7(0x07) | 40 |
| 1(0x01) | 8(0x08) | 30 |
| 1(0x01) | 9(0x09) | 20 |
| 1(0x01) | 10(0x0A) | 10 |
| 1(0x01) | 11(0x0B) | 5 |
| 1(0x01) | 12(0x0C) | 3 |
| 1(0x01) | 13(0x0D) | 2 |
| 1(0x01) | 14(0x0E) | 1 |